



Greedy Algorithm

Abdullah All Mamun Siam

Lecturer, Dhaka International University(DIU)

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Introduction to Greedy Algorithm

Greedy algorithms build up a solution step by step, always choosing the option that offers the most immediate benefit. Used for optimization problems where local choices lead to a globally optimal solution.

If an optimal solution to the problem can be found by choosing the best choice at each step without reconsidering the previous steps once chosen, the problem can be solved using a greedy approach. This property is called greedy choice property.

If both of the properties below are true, a greedy algorithm can be used to solve the problem.

Greedy choice property: A global (overall) optimal solution can be reached by choosing the optimal choice at each step.

Optimal substructure: A problem has an optimal substructure if an optimal solution to the entire problem contains the optimal solutions to the sub-problems.

Characteristics Of Greedy Algorithm

The key characteristics of a greedy algorithm:

1. Greedy choice property
2. Optimal substructure
3. No backtracking
4. Local optimization
5. Making decisions based on the best available option at each step

Example of Greedy Algorithm

Common examples:

1. Fractional Knapsack Problem
2. Activity Selection Problem
3. Huffman Encoding
4. Shortest Path Algorithm(Dijkstra's Algorithm)
5. Minimum Spanning Tree(Kruskal's and Prim's Algorithms)

Advantages

Advantages of Greedy Algorithms:

- 1.Simple and easy to understand
- 2.Efficient for certain problems
- 3.Fast execution
- 4.Can be used as building blocks for more complex algorithms

Disadvantages

Disadvantages of Greedy Algorithms:

1. Does not work for all optimization problems.
2. Can lead to suboptimal solutions in some cases.
3. Requires careful analysis before implementation.

Fractional Knapsack Problem

Given the weights and profits of N items, put these items in a knapsack of capacity W to get the maximum total profit in the knapsack.

In Fractional Knapsack, we can break items for maximizing the total value of the knapsack.

Fractional Knapsack Solution Using Greedy Approach

Step 1: Calculate the ratio (profit/weight) for each item.

Step 2: Sort all the items in decreasing order of the ratio.

Step 3: Do the following for every item i in the sorted order :

i.If the weight of the current item is less than or equal to the remaining capacity then add the value of that item into the result.

ii.Else add the current item as much as we can and break out of the loop.

Fractional Knapsack Solution Using Greedy Approach

Consider the example:

Profit={100,60,120}

Weight={20,10,30}

Capacity=50

Profit/Weight={5,6,4}

Sorted (Profit/Weight)={6,5,4}

Weight = 10 which is less than Capacity. So add this element in the knapsack. profit = 60 and remaining Capacity = $50 - 10 = 40$.

Weight = 20 which is less than Capacity. So add this element too. profit = $60 + 100 = 160$ and remaining Capacity = $40 - 20 = 20$.

Weight = 30 is greater than Capacity. So add $20/30$ fraction = $2/3$ fraction of the element. Therefore profit = $2/3 * 120 + 160 = 80 + 160 = 240$ and remaining Capacity becomes 0.

So the final profit becomes 240 for Capacity = 50.

Activity Selection Problem

You are given n activities with their start and finish times. Select the maximum number of activities that can be performed by a single person, assuming that a person can only work on a single activity at a time.

Activity Selection Problem Solution Using Greedy Approach

Step 1: Sort the activities according to their finishing time.

Step 2: Select the first activity from the sorted array.

Step 3: Do the following for the remaining activities in the sorted array

If the start time of this activity is greater than or equal to the finish time of the previously selected activity then select this activity.

Step 4: Repeat until all activities have been considered.

Activity Selection Problem Solution Using Greedy Approach

	0	1	2	3	4	5	6
START	1	3	2	0	5	8	11
END	3	4	5	7	9	10	12

SELECTED

START	1	3	2	0	5	8	11
END	3	4	5	7	9	10	12

START[1] > END[0], SELECTED

START	1	3	2	0	5	8	11
END	3	4	5	7	9	10	12

START[2] < END[1], REJECTED

START	1	3	2	0	5	8	11
END	3	4	5	7	9	10	12

START[3] < [1], REJECTED

So Selected Activity={0,1,4,6}

Thanks!!!